THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The perimeter of a square of side s is $4s$. For $s = 7$ cm it is (in cm): 1

(a) 28 **(b)** 29 **(c)** 27 **(d)** 30

2. The rule 'multiply by 5 and add 2' for the input n is: 1

(a) $5n + 2$ **(b)** $2n + 5$ **(c)** $5(n + 2)$ **(d)** $n + 7$

3. If $m = 4$, the value of $m + 3m$ is: 1

(a) 17 **(b)** 15 **(c)** 18 **(d)** 16

4. Subtract $4x$ from $9x$. The result is: 1

(a) $-5x$ **(b)** $5x$ **(c)** 5 **(d)** $13x$

5. The value of $2(x + 3)$ for $x = 4$ is: 1

(a) 15 **(b)** 14 **(c)** 13 **(d)** 16

6. 'Four times the sum of x and 3' is: 1

(a) $x + 12$ **(b)** $4 + 3x$ **(c)** $4(x + 3)$ **(d)** $4x + 3$

7. A taxi charges ₹50 fixed plus ₹12 per km. For $k = 5$ km the fare is (in ₹): 1

(a) 109 **(b)** 110 **(c)** 112 **(d)** 111

8. The number of matchsticks in a row of n squares is $3n + 1$. For $n = 6$ squares the number of matchsticks is: 1

(a) 20 **(b)** 21 **(c)** 18 **(d)** 19

9. The perimeter of a square of side s is $4s$. For $s = 12$ cm it is (in cm): 1

(a) 47 **(b)** 49 **(c)** 48 **(d)** 50

10. A rule gives 7, 12, 17, 22 for $n = 1, 2, 3, 4$. The rule is: 1

(a) $5n + 2$ **(b)** $5n$ **(c)** $7n$ **(d)** $n + 6$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The number of matchsticks needed for a row of n squares is $3n + 1$. How many matchsticks are needed for 5 squares? 2

12. Simplify: $7x + 4 - 3x + 6$. 2

13. Find the value of $5p - 3q$ for $p = 6$ and $q = 4$. 2

14. The number of matchsticks needed for a row of n squares is $3n + 1$. How many matchsticks are needed for 8 squares? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern, figure n has $3n + 2$ dots. Find the number of dots in figure 4 and in figure 5. How many more dots does figure 5 have? 3

16. Add $5x + 9y$ and $3x - 4y$. 3

17. A father's age is 3 times his son's age s, plus 2 years. Write an expression for the father's age and find it when $s = 10$. 3

OR

In a pattern, figure n has $3n + 2$ dots. Find the number of dots in figure 6 and in figure 7. How many more dots does figure 7 have? 3

18. Add $7x + 6y$ and $2x - y$. 3

OR

A father's age is 2 times his son's age s, plus 8 years. Write an expression for the father's age and find it when $s = 14$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A school buys n notebooks at ₹12 each and 4 pens at ₹5 each. Write an expression for the total cost. Find the cost when $n = 10$ and the change from ₹200. 5

OR

A hall has 8 rows of r chairs and 6 extra chairs at the back. Write an expression for the total number of chairs and find it for $r = 12$ and for $r = 14$. 5

20. A school buys n notebooks at ₹15 each and 3 pens at ₹8 each. Write an expression for the total cost. Find the cost when $n = 8$ and the change from ₹200. 5

OR

A hall has 10 rows of r chairs and 4 extra chairs at the back. Write an expression for the total number of chairs and find it for $r = 9$ and for $r = 11$. 5

END OF QUESTION PAPER — Total: 40 Marks

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